

QF4102 Financial Modelling and Computation

Assignment 1

AY26/27 Semester 1

1. Under the Black-Scholes model, there are a series of closed-form solutions for Barrier options. In particular, from Hull 2012, Section 25.8, the price of a down-and-in European call option when $H \leq X$ is given by

$$c_{di} = S_0 e^{-qT} (H/S_0)^{2\lambda} N(y) - X e^{-rT} (H/S_0)^{2\lambda-2} N(y - \sigma\sqrt{T}),$$

where

$$\lambda = \frac{r - q + \sigma^2/2}{\sigma^2},$$
$$y = \frac{\ln[H^2/(S_0 X)]}{\sigma\sqrt{T}} + \lambda\sigma\sqrt{T}.$$

- (i) Implement the Black-Scholes formula for a down-and-in barrier call option, using the function header

```
def bs_diCall(S0,X,r,T,sigma,H,q=0):  
    pass
```

Your function should allow for the input arguments `S0` and `H` to be scalars or 1D arrays of compatible sizes.

- (ii) A down-and-in call option has a time to maturity of 1 year, where the strike price is \$1.4, volatility of the underlier is 20%, the underlying asset pays a continuous dividend yield of 1%, and where the risk free rate is 4%.
Use your function to compute the option values with $H = \$1.35$, and with S_0 on a grid with the range \$1.41 to \$1.7, in increments of \$0.01. Plot the option values versus S_0 ; and comment on the graph obtained.
- (iii) For the same range of S_0 , add in the Black-Scholes prices for a vanilla European call onto the graph in (ii), and comment.
- (iv) For the same set of parameters in (ii), compute the value of the option with $S_0 = \$1.45$, and with the barrier H over the range from \$1.1 to \$1.4, in increments of \$0.01. Plot the option values versus H ; and comment on the graph obtained. Add in the Black-Scholes vanilla European call price to the graph in (iv), and comment.

2. Barrier options satisfy an in-out parity relation. In particular, if c_{do} and c_{di} are the call prices for the knock-in and knock-out counterparts respectively, with the same terms, then

$$c_{do} + c_{di} = c_{\text{vanilla}}.$$

- (i) Using the above parity relation, implement a function to estimate the price of a down-and-in call option using the **binomial model**. Use the header

```
def bopm_diCall(S0,X,r,T,sigma,H,N,q=0):
    pass
```

- (ii) Using the same parameters in question 1(iv), make use of your function to estimate the option values when the barrier is at \$1.35, for $N = 1700, 1701, \dots, 2200$. Taking the Black-Scholes price to be the true solution, obtain the absolute errors of your estimates, and plot these versus N . Comment on the graph obtained.
- (iii) Which values of N correspond to the 2 local minimum absolute errors? For these values of N , determine the corresponding values of i for the artificial barrier. Include all relevant workings in your report.
3. (i) For the floating-strike lookback **call** option (that is, a lookback on minimum), a reduction in variables can be applied to reduce the two state variables A, S to a single state variable $x = \ln(A/S)$, where $V(S, A, t) = SW(x, t)$.
Provide an algorithm which uses the single-state-variable x to price an **American** floating strike lookback *call* option using the BOPM. Assume that the option is *newly issued*.
- (ii) Implement the algorithm as a function with the header
- ```
def bopm1Vx_flLookAmCall(S0,r,T,sigma,N,q=0):
 pass
```
- (iii) Use your function in (ii) to obtain estimates of the value of a newly issued American floating strike lookback call. The option has 6 months to maturity, with a current underlier price of \$1.25, and a volatility of 0.22. The underlying asset pays a continuous dividend yield of 3%, and the risk free rate is 3%.  
Obtain estimates where the number of time steps  $N$  ranges from 200 to 4000, in increments of 200. Plot the option values versus  $N$ ; and comment on the graph obtained.
- (iv) Obtain estimates for an European option with the same parameters in (iii), using the same range of values for  $N$ , but without the possibility of early exercise. Plot the option values versus  $N$ , and include the values from (iii) on the same plot. Compare your results and comment.

# Submission

Submit a .zip folder to Canvas by **19 September 2026, 2359h** containing the following:

1. A pdf report. This report should contain all results, workings, supporting figures, tables, as well as analysis and comments for the assignment.
2. Supporting python scripts — each of these scripts may contain functions as required by the assignment, or functions that support the main scripts.
3. Main python scripts — For **each main question** in the assignment, submit one script containing all code that reproduces your results and figures. (The main scripts may import functions from other supporting scripts — refer to point 2.)
  - Name these files `assm1_q1.py`, `assm1_q2.py` and `assm1_q3.py` respectively.
  - As each of these scripts will be run during the grading process, do ensure that each script is executable and runs without errors.
  - If a supporting function is required, import the function in your main script(s), and include its containing file in your submission.

*Note:* Students may discuss the assignment with each other, but each each group must submit their own work. In the event of plagiarism or a violation of the academic integrity policy, all parties involved will be awarded zero for this assignment.